

End of Topic Test 2 — Mark Scheme

Chapter 7: Correlation and Regression

Total: 58 marks

Notation: M1 = method mark; A1 = accuracy mark (dependent on correct method); B1 = independent mark for a correct statement, value or comment. Accept equivalent correct methods and equivalent valid comments/context unless otherwise stated.

Question 1

[18 marks]

Ice cream sales and temperature. Data: x (temperature, °C): 14, 18, 21, 25, 28, 30, 17, 22, 26, 19. y (ice creams sold): 40, 62, 75, 95, 110, 125, 55, 80, 100, 65. ($n = 10$, $\bar{x} = 22.0$, $\bar{y} = 80.7$.)

- (a) M1 correct scales chosen on both axes, covering the data range. A1 at least 8 of the 10 points plotted correctly (allow ± 1 small square). A1 all 10 points plotted correctly, with no line drawn. [3]
- (b) B1 positive direction (as temperature increases, ice cream sales increase). B1 strong linear relationship. B1 contextual comment, e.g. warmer days lead to greater demand for ice cream, which is a sensible real-world relationship. [3]
- (c) M1 correct method/formula for r used (by calculator, or from summary statistics). M1 $S_{xx} = \sum(x - \bar{x})^2 = 240$. M1 $S_{yy} = \sum(y - \bar{y})^2 = 6344.1$. M1 $S_{xy} = \sum(x - \bar{x})(y - \bar{y}) = 1232$. A1 $r = \frac{1232}{\sqrt{240 \times 6344.1}} = 0.998$ (3 d.p.). B1 interpretation: a very strong, positive correlation between temperature and ice cream sales. [6]
- (d) M1 gradient: $b = S_{xy}/S_{xx} = 1232/240 = 5.13$ (2 d.p.). M1 intercept: $a = \bar{y} - b\bar{x} = 80.7 - 5.1333(22) = -32.23$, so $\hat{y} = -32.23 + 5.13x$. A1 substitute $x = 24$: $\hat{y} \approx -32.23 + 5.13(24) \approx 91$ ice creams. B1 this is interpolation, since $x = 24$ lies within the range of the given data ($14 \leq x \leq 30$). [4]
- (e) B1 reason 1, e.g. other factors affect daily sales independently of temperature (day of the week, whether it is raining, local events, price changes, promotions). B1 reason 2, e.g. natural random variation between days, or measurement/recording error, not captured by a simple linear model. [2]

Question 2

[14 marks]

Screen time, sleep and stress. $r(\text{screen time, sleep}) = -0.82$; $r(\text{screen time, stress}) = 0.65$; $n = 12$.

- (a) B1 negative direction. B1 strong correlation (magnitude close to 1). B1 contextual interpretation: students with more screen time tend to get less sleep. [3]
- (b) B1 positive direction. B1 moderate/fairly strong correlation. B1 contextual interpretation: students with more screen time tend to report higher stress. [3]

- (c) B1 the screen time–sleep relationship is stronger. B1 correct reason: $|-0.82| > |0.65|$, i.e. the magnitude of r (ignoring sign) is greater for screen time and sleep. [2]
- (d) B1 correlation does not imply causation – a correlation coefficient only measures association/strength of a linear pattern, not that one variable causes changes in the other. B1 alternative direction of causation possible, e.g. higher stress could lead students to use screens more (as a distraction or coping mechanism), rather than screen time causing the stress. B1 confounding variable suggested with justification, e.g. academic/exam workload could increase both screen time (used for revision or as a distraction from pressure) and stress levels independently. B1 overall clarity and quality of the explanation. [4]
- (e) B1 suitable study design suggested, e.g. a controlled experiment where screen time is deliberately varied between groups, or a longitudinal study tracking the same students' screen time and stress over time. B1 justification, e.g. this would help to rule out reverse causation and confounding variables, and show whether a change in screen time precedes a change in stress. [2]

Question 3

[14 marks]

Independent study and GCSE science scores. School P: $\hat{y} = 38 + 3.1x$; School Q: $\hat{y} = 45 + 2.4x$.

- (a) B1 School P intercept: 38 is the predicted score for a student who does 0 hours of independent study per week. B1 School P gradient: each extra hour of independent study is associated with an increase of 3.1 marks in predicted score. B1 School Q intercept: 45 is the predicted baseline score with 0 hours of study. B1 School Q gradient: each extra hour is associated with an increase of 2.4 marks in predicted score. [4]
- (b) B1 School P has the greater gradient ($3.1 > 2.4$). B1 correct comparison clearly stated. B1 meaning: each additional hour of independent study is associated with a bigger increase in predicted score at School P than at School Q. [3]
- (c) M1 School P prediction: $\hat{y} = 38 + 3.1(8) = 62.8$. M1 School Q prediction: $\hat{y} = 45 + 2.4(8) = 64.2$. A1 both residuals correct: School P = $70 - 62.8 = +7.2$; School Q = $70 - 64.2 = +5.8$. A1 comment: School Q's model fits this student's result more closely, since it has the smaller residual ($+5.8 < +7.2$). [4]
- (d) B1 reason 1, e.g. cohort/intake ability may differ between schools – School Q's higher intercept could reflect a stronger starting ability among its students, independent of study hours. B1 reason 2, e.g. teaching quality or the effectiveness of independent study resources may differ – School P's steeper gradient suggests each hour of study is more productive there. B1 overall quality/coherence of the explanation (e.g. also noting these are only two of several possible differences between the school samples). [3]

Question 4

[12 marks]

Wingspan and body mass in birds. $\hat{y} = -12 + 3.4x$, $r = 0.93$, $n = 9$, wingspan range 22–95 cm.

- (a) M1 substitute $x = 250$: $\hat{y} = -12 + 3.4(250)$. A1 $\hat{y} \approx 838$ g. B1 comment: $x = 250$ lies far outside the range of the sample data (22–95 cm), so this is extrapolation and the prediction is

unreliable/should not be trusted. [3]

- (b) B1 the large species is a high-leverage outlier, positioned far from the other 8 points in both x and y . B1 this inflates the calculated value of r , making the overall correlation look stronger than it is for typical-sized species (removing it would likely give a smaller, more moderate r). B1 it also has a strong influence (leverage) on the fitted regression line, pulling it towards this one point, so the line may not represent the relationship well for other, more typical species. [3]
- (c) B1+B1 any two valid limitations, e.g. the true relationship may not be linear across all species/sizes; wingspan alone does not account for differences in body shape, density or build between species; the small sample size ($n = 9$) means the fitted line may not be reliable or generalisable to other species. B1 development/clarity, e.g. explaining the impact of one limitation more fully in context. [3]
- (d) B1 a strong correlation shows association, not causation – $r = 0.93$ does not prove that a longer wingspan directly causes greater mass. B1 alternative explanation given, e.g. both wingspan and mass are likely determined together by a species' overall body size/genetics (larger species tend to evolve both larger wings and greater mass), rather than one causing the other. B1 clarity and quality of the explanation. [3]

END OF MARK SCHEME